Bayesian Inference

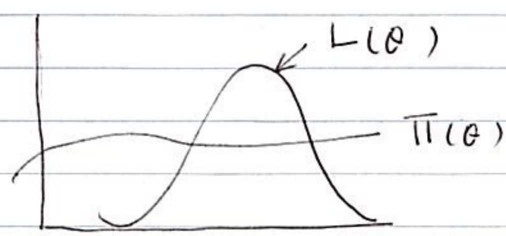
Frame works :

1) specify a prior $\pi(\theta)$, capture rough knowledge about θ (a r.v.)

2) specify a probability model $P(y_1, \cdots, y_n \mid \theta)$

3) Find the posterior

$$P(\theta \mid y_1, \cdots y_n) = \frac{P(y_1 \cdots y_n \mid \theta)\,\pi(\theta)}{\int P(y_1 \cdots y_n \mid \theta)\,\pi(\theta)\,d\theta}$$



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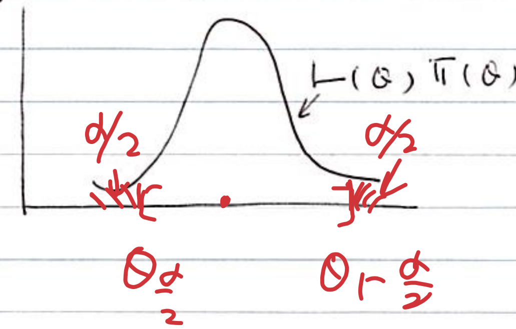


4) Interpretation

A) Summary θ :

1) $L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$; $\hat{\theta} = E(\theta \mid y_1, \cdots y_n)$

2) $L(\theta, \hat{\theta}) = |\theta - \hat{\theta}|$; $\hat{\theta} = \text{median}(\theta \mid y_1, \cdots y_n)$

$$\hat{\theta} = \underset{\hat{\theta}}{\arg\min}\; E(L(\theta, \hat{\theta}) \mid y_1, \cdots y_n)$$

3) $L(\theta, \hat{\theta}) = I(\theta \neq \hat{\theta})$

$\hat{\theta} = \text{mode of } \theta \mid y_1, \cdots y_n$

5) Made prediction

$$P(y_{n+1} \mid y_1, \cdots y_n)$$

$$= \frac{\int P(y_{n+1}, y_1, \cdots y_n \mid \theta)\, \pi(\theta)\, d\theta}{\int P(y_1, \cdots y_n \mid \theta)\, \pi(\theta)\, d\theta}$$

· If, we assum

$y_1, y_2, \cdots y_n, y_{n+1} \mid \theta$ are independent

then, $P(y_{n+1} \mid y_1, \cdots y_n)$

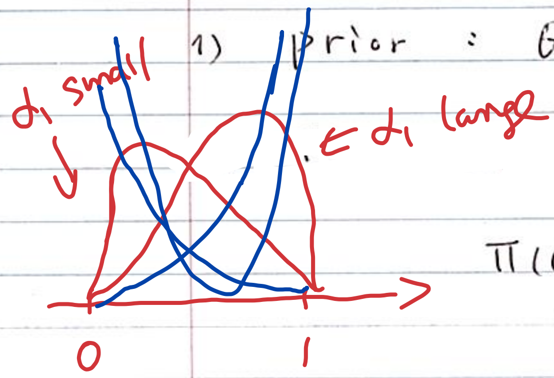
$$= \int P(y_{n+1} \mid \theta)\, P(\theta \mid y_1, \cdots y_n)\, d\theta$$

$$\approx \Sigma P(y_{n+1} \mid \theta_j) \,/\, J$$

example:

$$y_1, y_2, \cdots y_n \mid \theta \overset{iid}{\sim} \text{Bern}(\theta)$$

1) prior : $\theta \sim \text{Beta}(\alpha_1, \alpha_0)$ → given



$$\pi(\theta) = \frac{\Gamma(\alpha_0+\alpha_1)}{\Gamma(\alpha_0)\cdot\Gamma(\alpha_1)}\,\theta^{\alpha_1-1}\cdot(1-\theta)^{\alpha_0-1}$$

e.g. Beta(1, 1) = Unif([0,1])

Beta(2, 1)

$$E(\theta) = \frac{\alpha_1}{\alpha_0+\alpha_1}$$

$$V(\theta) = \frac{\alpha_0\alpha_1}{(\alpha_1+\alpha_0)^2(\alpha_0+\alpha_1+1)}$$

3) likelihood :

$$P(y_i \mid \theta) = \theta^{y_i}(1-\theta)^{1-y_i}$$

$$P(y_1, \cdots y_n \mid \theta) = \prod_{i=1}^{n} P(y_i \mid \theta)$$

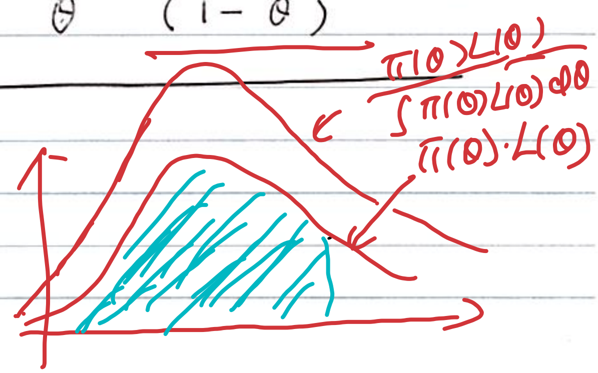
$$= \theta^{\Sigma y_i}(1-\theta)^{n-\Sigma y_i}$$

Let $\underset{\sim}{y} = (y_1, \cdots y_n)$

4) Posterior :

$$P(\theta \mid y_1, \cdots, y_n) = \frac{\pi(\theta)\,P(\underset{\sim}{y} \mid \theta)}{P(y_1, \cdots y_n)}$$

$$= \frac{\frac{\Gamma(\alpha_0+\alpha_1)}{\Gamma(\alpha_0)\cdot\Gamma(\alpha_1)}\,\theta^{\alpha_1-1}\underline{(1-\theta)^{\alpha_0-1}}\;\theta^{\Sigma y_i}\underline{(1-\theta)^{n-\Sigma y_i}}}{P(\underset{\sim}{y})}$$



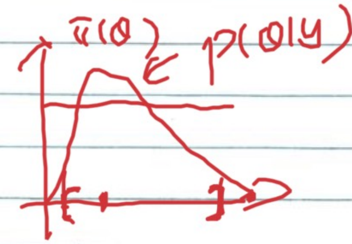
$$\propto \theta^{\overbrace{\alpha_1 + \Sigma y_i}^{\alpha_1'} - 1} (1-\theta)^{\overbrace{\alpha_0 + n - \Sigma y_i}^{\alpha_0'} - 1}$$

We see that

★ $\theta \mid \underset{\sim}{y} \sim \text{Bata}(\alpha_1 + n_1, \ \alpha_0 + n_0)$

where, $n_1 = \Sigma y_i \qquad n_0 = n - \Sigma y_i$

$$E(\theta \mid \underset{\sim}{y}) = \frac{\alpha_1 + n_1}{\alpha_0 + \alpha_1 + n} \to \theta$$



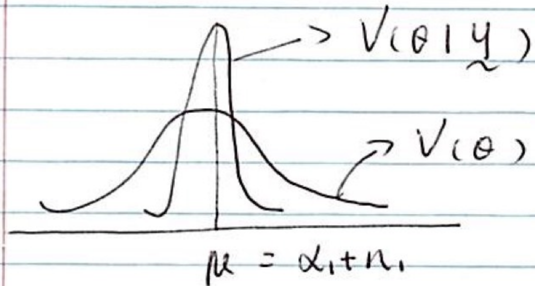
$$\hat{\theta}_{mle} = \frac{n_1}{n}$$

? $$V(\theta \mid \underset{\sim}{y}) = \frac{(\alpha_1 + n_1)(\alpha_0 + n_0)}{(n + \alpha_0 + \alpha_1)^2 (n + \alpha_0 + \alpha_1 - 1)} = O\left(\frac{1}{n}\right)$$

$V(\theta \mid \underset{\sim}{y})$ can converge to zero

when as $n \to \infty$

$$\hat{\theta}_{mle} \sim N\left(\theta_0, \frac{1}{n I(\theta)}\right)$$



$\Rightarrow$ Generaly :

(1) when we need to draw sample from $P(\theta | y) \propto \pi(\theta) P(y | \theta)$

2), We need to find $E(a(\theta) | \underset{\sim}{y})$

$$= \int a(\theta) P(\theta | \underset{\sim}{y}) d\theta$$